

Question # 1: Multiple choice question. [15 x 1pt]

1- An example on stack application:

- a) Navigating web pages
- b) Serving requests on a printer
- c) Handling of interrupts in real-time systems
- d) Process scheduling in operating systems

2- Using stack, what is the postfix expression of the infix expression: $((A * (B - C)) - D) / (E + F)$.

- a) $ABC * - D - E + F /$
- b) $AB - C * D - EF / +$
- c) $ABC - D * - EF + /$
- d) $ABC - * D - EF + /$

3- If the SIZE is the size of the array used in the implementation of the circular queue. How is the rear variable manipulated while inserting an element in the queue?

- a) $rear = (rear \% 1) + SIZE$
- b) $rear = rear + (1 \% SIZE)$
- c) $rear = rear \% (SIZE + 1)$
- d) $rear = (rear + 1) \% SIZE$

4- How do you determine if the queue is empty?

- a) $front == rear$
- b) $front == rear - 1$
- c) $front == - 1$
- d) $front == rear + 1$

5- What is the primary advantage of a circular queue over a linear queue?

- a) Dynamic resizing
- b) Constant-time searching
- c) Reusing of space left by previously dequeued elements
- d) Simpler implementation

6- Which of the following function is **not** required in the linked list implementation of a stack?

- a) `isEmpty()`
- b) `isFull`
- c) `() push()`
- d) `pop()`
- e) None

7- In the linked list implementation of a stack, where does a new element be inserted?

- a) At the head of the link list.
- b) At the tail of the link list.
- c) Depending on the index value stored in `tos` variable
- d) Depending on the index value stored in `rear` variable

8- Which of the following operation is quite difficult in an array when compared to linked lists?

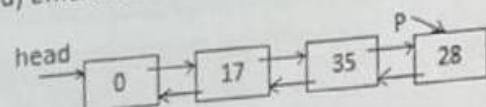
- a) Accessing elements
- b) Deleting elements at the end
- c) Inserting elements in the middle
- d) Traversing

9- Which of the following is an advantage of using linked lists over arrays?

- a) Dynamic size
- b) Higher memory overhead due to pointers
- c) Sequential memory allocation
- d) Efficient traversal

10- What is the expression to access to the value 35 in the following doubly linked list?

- a) `head->next->info;`
- b) `head->next->next->info;`
- c) `p->back;`
- d) `p->back->next->info;`
- e) `p->back->back->info;`
- f) `p->back->back;`



11- What happens if you insert elements in sorted order into a binary search tree?

- a) The tree becomes balanced
- b) The height remains logarithmic
- c) The tree becomes a linked list
- d) a & b

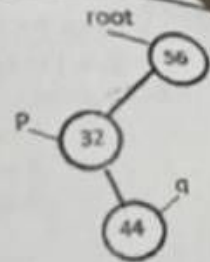
12- In order to get the information stored in a binary search tree in the descending order, one should traverse it in which of the following order?

- a) Left, root, right
- b) root, left, right
- c) right, left, root
- d) right, root, left



13- How do you delete the node with the value 56 from the following BST?

- a) root->info = q->info; p->right = NULL; delete q;
- b) root->info = q->info; root->left = q; delete q;
- c) root->info = p->info; root->left = q; delete p;
- d) root->info = p->info; p->right = NULL; delete p;



d) None

14- Which traversal method is used in BSTs to delete all nodes in a tree safely?

- a) Preorder
- b) Inorder
- c) Postorder

15- What will happen if a recursive function doesn't have a proper base case?

- a) The function will terminate immediately
- b) The function will continue running indefinitely, causing a stack overflow
- c) The function will return an incorrect result
- d) The function will throw a compile-time error

Question # 2: Fill in the blanks. [10 x 1pts]

16- What is the result of the following postfix expressions? $4\ 5\ 2\ -\ * \ 3\ -\ 1\ 2\ +\ /\ =$ _____

17- A circular queue is implemented using an array of size 9. The array index starts with 0, the rear is 6, and the front is 8. At which array index does the deletion of the next element take place? _____

18- How many temporary pointers are needed to delete a node in singly linked list? _____

19- How many pointers are there in doubly linked list must be updated to delete a node in the middle? _____

20- What is the right node of the root node in the following postorder traversal of a binary tree nodes: _____

C, K, A, H, B, G, D, F, E _____

21- How many internal nodes are there in a full binary tree with a total of 31 nodes? _____

22- How many leaves are there in a full binary tree of height equals to 6? _____

23- How many pointers are there in a binary tree must be updated to delete a node has no children? _____

24- How many pointers are there in a binary tree must be updated to delete a node has 2 children? _____

25- How many function calls are required to calculate Fibonacci numbers for fib(3)? _____

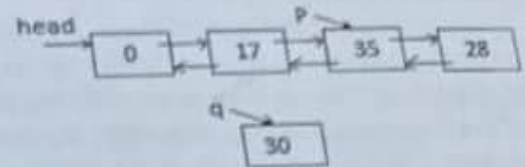
*** Write the correct answers of the Questions # 1 & # 2 in the table below:

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

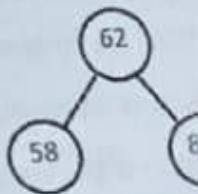
Question # 3: [5 pts]

a) Redraw the following linked list after executing the following code:

```
head->next = p;
q->next = p->next ;
p->next->back = q;
p->back = head;
q->back = p;
p->next = q;
```



b) Redraw the following binary search tree after executing the following code:



<pre>BinarySearchTree<int> bst; bst.insert(70); bst.insert(65); bst.insert(80); bst.delete_node(58);</pre>	<pre>BinarySearchTree<int> bst; bst.insert(70); bst.insert(65); bst.insert(80); bst.delete_node(58); bst.delete_node(88);</pre>	<pre>BinarySearchTree<int> bst; bst.insert(70); bst.insert(65); bst.insert(80); bst.delete_node(58); bst.delete_node(88); bst.delete_node(62);</pre>